Aim: To perform Non-Graphical Exploratory Data Analysis.

What is Mean?

The mean (or arithmetic average) is a measure of central tendency calculated by summing all values in a dataset and dividing by the total number of values.

Formula: (Sum of values / Number of values)

What is Median?

The median is the middle value in an ordered dataset (arranged from lowest to highest), representing the 50th percentile where half the data points are below it and half are above.

- **Odd Number of Data Points:** The median is the exact middle number.
- **Even Number of Data Points:** The median is the average (mean) of the two middle numbers.

What is Mode?

The mode is the value that appears most frequently in a data set, serving as a measure of central tendency to identify the most common item or "peak" in data.

Types:

- Unimodal: One value occurs most frequently.
- Bimodal/Multimodal: Two or more values tie for the highest frequency.
- No Mode: No values repeat.

What is Standard Deviation?

A standard deviation (or σ) is a measure of how dispersed the data is in relation to the mean. Low, or small, standard deviation indicates data are clustered tightly around the mean, and high, or large, standard deviation indicates data are more spread out.

$$S = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$$

What is Variance?

Variance is a statistical measure of how far a set of data points are spread out from their average (mean) value. Defined as the average of the squared differences from the mean, a low variance indicates data is clustered near the mean, while a high variance indicates data is widely dispersed.

$$\sigma^2 = \frac{\sum(xi - \bar{x})^2}{N}$$

What is IQR?

The Interquartile Range (IQR) is a measure of statistical dispersion that represents the spread of the middle 50% of a data set, calculated by subtracting first quartile Q1 from third Q3.

Formula: IQR = Q3 – Q1

What is Kurtosis?

Kurtosis is a statistical measure that defines the heaviness of the tails of a probability distribution relative to a normal distribution. It indicates the frequency of outliers (extreme values) in a dataset, rather than just the peak's height. High kurtosis means more, extreme outliers, while low kurtosis indicates fewer, lighter outliers.

Types of Kurtosis:

- Leptokurtic (High Kurtosis): Sharp peak, heavy tails, high outlier risk.
- Mesokurtic (Normal Kurtosis): Moderate, normal distribution shape.
- Platykurtic (Low Kurtosis): Flat peak, thin tails, low outlier risk.

Dataset:

id	name	developer	genre	release_year	metacritic_score	sales_millions	peak_players
1	Call of Duty 4: Modern Warfare	Infinity Ward	First-Person Shooter	2007	94	15.7	50000
2	Call of Duty: Modern Warfare 2	Infinity Ward	First-Person Shooter	2009	94	25	120000
3	Call of Duty: Black Ops	Treyarch	First-Person Shooter	2010	87	30.7	150000
4	Call of Duty: Warzone	Infinity Ward	Battle Royale	2020	80	0	6000000
5	Call of Duty: Modern Warfare (2019)	Infinity Ward	First-Person Shooter	2019	80	41	500000
6	Resident Evil 2	Capcom	Survival Horror	1998	93	4.96	0
7	Resident Evil 4	Capcom	Survival Horror	2005	96	7.1	0
8	Resident Evil 7: Biohazard	Capcom	Survival Horror	2017	86	12.4	20000
9	Resident Evil Village	Capcom	Survival Horror	2021	84	8.5	100000
10	Resident Evil (Remake)	Capcom	Survival Horror	2002	91	1.35	0
11	Grand Theft Auto V	Rockstar North	Action-Adventure	2013	97	195	500000
12	Red Dead Redemption 2	Rockstar Studios	Action-Adventure	2018	97	65	120000
13	Grand Theft Auto: San Andreas	Rockstar North	Action-Adventure	2004	93	27.5	0
14	Bully	Rockstar Vancouver	Action-Adventure	2006	87	1.5	0
15	Max Payne 3	Rockstar Studios	Third-Person Shooter	2012	87	4	5000
16	God of War	Santa Monica Studio	Action-Adventure	2018	94	23	0
17	The Last of Us	Naughty Dog	Action-Adventure	2013	95	20	0
18	Uncharted 4: A Thief's End	Naughty Dog	Action-Adventure	2016	93	16	0
19	Spider-Man	Insomniac Games	Action-Adventure	2018	87	20	0
20	Horizon Zero Dawn	Guerrilla Games	Action RPG	2017	89	20	0

Code:

```
import pandas as pd

df = pd.DataFrame(dataset)

print()
print("=" * 122)
print(f"{'ID':<4} {'Name':<35} {'Developer':<20} {'Genre':<18} {'Year':<6}
{'Metacritic':<10} {'Sales (M)':<10} {'Peak Players':<12}")
print("-" * 122)

for game in dataset:
    print(f"{game['id']:<4} {game['name'][:34]:<35} {game['developer'][:19]:<20}
{game['genre'][:17]:<18} "
          f"{game['release_year']:<6} {game['metacritic_score']:<10}
{game['sales_millions']:<10.1f} {game['peak_players']:<12,}")

print("=" * 122)
```

```
mean_metacritic_score = df["metacritic_score"].mean()
mean_sales = df["sales_millions"].mean()
mean_peak_players = df["peak_players"].mean()

print(f"\nAverage Metacritic Score: {mean_metacritic_score:.2f}")
print(f"Average Sales (Millions): {mean_sales:.2f}")
print(f"Average Peak Players: {mean_peak_players:.2f}")

median_metacritic_score = df["metacritic_score"].median()
median_sales = df["sales_millions"].median()

print(f"\nMedian Metacritic Score: {median_metacritic_score:.2f}")
print(f"Median Sales (Millions): {median_sales:.2f}")

mode_metacritic_score = df["metacritic_score"].mode()[0]
mode_sales = df["sales_millions"].mode()[0]

print(f"\nMode Metacritic Score: {mode_metacritic_score:.2f}")
print(f"Mode Sales (Millions): {mode_sales:.2f}")

std_metacritic_score = df["metacritic_score"].std()
std_sales = df["sales_millions"].std()

print(f"\nStandard Deviation of Metacritic Score: {std_metacritic_score:.2f}")
print(f"Standard Deviation of Sales (Millions): {std_sales:.2f}")

variance_metacritic_score = df["metacritic_score"].var()
variance_sales = df["sales_millions"].var()

print(f"\nVariance of Metacritic Score: {variance_metacritic_score:.2f}")
print(f"Variance of Sales (Millions): {variance_sales:.2f}")

quartile1_metacritic_score = df["metacritic_score"].quantile(0.25)
quartile3_metacritic_score = df["metacritic_score"].quantile(0.75)

print(f"\n1st Quartile of Metacritic Score: {quartile1_metacritic_score:.2f}")
print(f"3rd Quartile of Metacritic Score: {quartile3_metacritic_score:.2f}")

kurtosis_metacritic_score = df["metacritic_score"].kurtosis()
kurtosis_sales = df["sales_millions"].kurtosis()

print(f"\nKurtosis of Metacritic Score: {kurtosis_metacritic_score:.2f}")
print(f"Kurtosis of Sales (Millions): {kurtosis_sales:.2f}")
```

Output:

```
$ py app.py

==========================================================================================================================
ID   Name                                  Developer            Genre               Year   Metacritic Sales (M)  Peak Players
--------------------------------------------------------------------------------------------------------------------------
1    Call of Duty 4: Modern Warfare        Infinity Ward        First-Person Shoo  2007   94         15.7       50,000
2    Call of Duty: Modern Warfare 2        Infinity Ward        First-Person Shoo  2009   94         25.0       120,000
3    Call of Duty: Black Ops               Treyarch             First-Person Shoo  2010   87         30.7       150,000
4    Call of Duty: Warzone                 Infinity Ward        Battle Royale      2020   80         0.0        6,000,000
5    Call of Duty: Modern Warfare (2019    Infinity Ward        First-Person Shoo  2019   80         41.0       500,000
6    Resident Evil 2                       Capcom               Survival Horror    1998   93         5.0        0
7    Resident Evil 4                       Capcom               Survival Horror    2005   96         7.1        0
8    Resident Evil 7: Biohazard            Capcom               Survival Horror    2017   86         12.4       20,000
9    Resident Evil Village                 Capcom               Survival Horror    2021   84         8.5        100,000
10   Resident Evil (Remake)                Capcom               Survival Horror    2002   91         1.4        0
11   Grand Theft Auto V                    Rockstar North       Action-Adventure   2013   97         195.0      500,000
12   Red Dead Redemption 2                 Rockstar Studios     Action-Adventure   2018   97         65.0       120,000
13   Grand Theft Auto: San Andreas         Rockstar North       Action-Adventure   2004   93         27.5       0
14   Bully                                 Rockstar Vancouver   Action-Adventure   2006   87         1.5        0
15   Max Payne 3                           Rockstar Studios     Third-Person Shoo  2012   87         4.0        5,000
16   God of War                            Santa Monica Studio  Action-Adventure   2018   94         23.0       0
17   The Last of Us                        Naughty Dog          Action-Adventure   2013   95         20.0       0
18   Uncharted 4: A Thief's End            Naughty Dog          Action-Adventure   2016   93         16.0       0
19   Spider-Man                            Insomniac Games      Action-Adventure   2018   87         20.0       0
20   Horizon Zero Dawn                     Guerrilla Games      Action RPG         2017   89         20.0       0
==========================================================================================================================

Average Metacritic Score: 90.20
Average Sales (Millions): 26.94
Average Peak Players: 378250.00

Median Metacritic Score: 92.00
Median Sales (Millions): 18.00

Mode Metacritic Score: 87.00
Mode Sales (Millions): 20.00

Standard Deviation of Metacritic Score: 5.25
Standard Deviation of Sales (Millions): 42.49

Variance of Metacritic Score: 27.54
Variance of Sales (Millions): 1805.10

1st Quartile of Metacritic Score: 87.00
3rd Quartile of Metacritic Score: 94.00

Kurtosis of Metacritic Score: -0.62
Kurtosis of Sales (Millions): 14.25
```

Conclusion

Successfully demonstrated non-graphical data analysis using pandas for the given dataset.

```
skewness_metacritic_score = df["metacritic_score"].skew()
skewness_sales = df["sales_millions"].skew()

print(f"\nSkewness of Metacritic Score: {skewness_metacritic_score:.2f}")
print(f"Skewness of Sales (Millions): {skewness_sales:.2f}")

if abs(skewness_metacritic_score) < 0.5:
    print(f"\nMetacritic Score: Approximately symmetric (skewness =
{skewness_metacritic_score:.2f})")
elif skewness_metacritic_score < 0:
    print(f"Metacritic Score: Negatively skewed (left-skewed) =
{skewness_metacritic_score:.2f}")
else:
    print(f"Metacritic Score: Positively skewed (right-skewed) =
{skewness_metacritic_score:.2f}")

if abs(skewness_sales) < 0.5:
    print(f"Sales: Approximately symmetric (skewness = {skewness_sales:.2f})")
elif skewness_sales < 0:
    print(f"Sales: Negatively skewed (left-skewed) = {skewness_sales:.2f}")
else:
    print(f"Sales: Positively skewed (right-skewed) = {skewness_sales:.2f}")
```

```
Average Metacritic Score: 90.20
Average Sales (Millions): 26.94
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Mode Metacritic Score: 87.00
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Standard Deviation of Metacritic Score: 5.25
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Variance of Metacritic Score: 27.54
Variance of Sales (Millions): 1805.10

1st Quartile of Metacritic Score: 87.00
3rd Quartile of Metacritic Score: 94.00

Kurtosis of Metacritic Score: -0.62
Kurtosis of Sales (Millions): 14.25

Skewness of Metacritic Score: -0.57
Skewness of Sales (Millions): 3.59
Metacritic Score: Negatively skewed (left-skewed) = -0.57
Sales: Positively skewed (right-skewed) = 3.59
```